potwatch

Know when a Portaldot network is unhealthy, why it is unhealthy, and prove it on-chain.

Portaldot Hacker House 2026 · Node Operations & Tooling

The problem

- A dApp developer sees "my transaction never confirms". The cause sits several layers down: a node with 0 peers, an offline GRANDPA voter, or a stalled indexer.
- Portaldot V3 has three layers to watch: Substrate (BABE + GRANDPA), the EVM, and the explorer.
- Today, checking them takes polkadot.js, an EVM wallet, the explorer, and knowledge of consensus internals.
- There is also no public, verifiable record of network uptime that builders and operators can point to.

What potwatch does

- One command checks every layer: `python -m potwatch check`
- Decodes BABE slots and authors from block headers, plus GRANDPA rounds, validators and session keys
- 23 finding types, each with a plain-language hint
- Output as text, HTML, Prometheus metrics, and an exit code for cron or CI
- Read-only by construction: an RPC method allowlist, and no keys
- Cross-network check: flags keys reused between networks
- `watch` mode logs state changes
- Rules are pure functions, tested against real captured snapshots

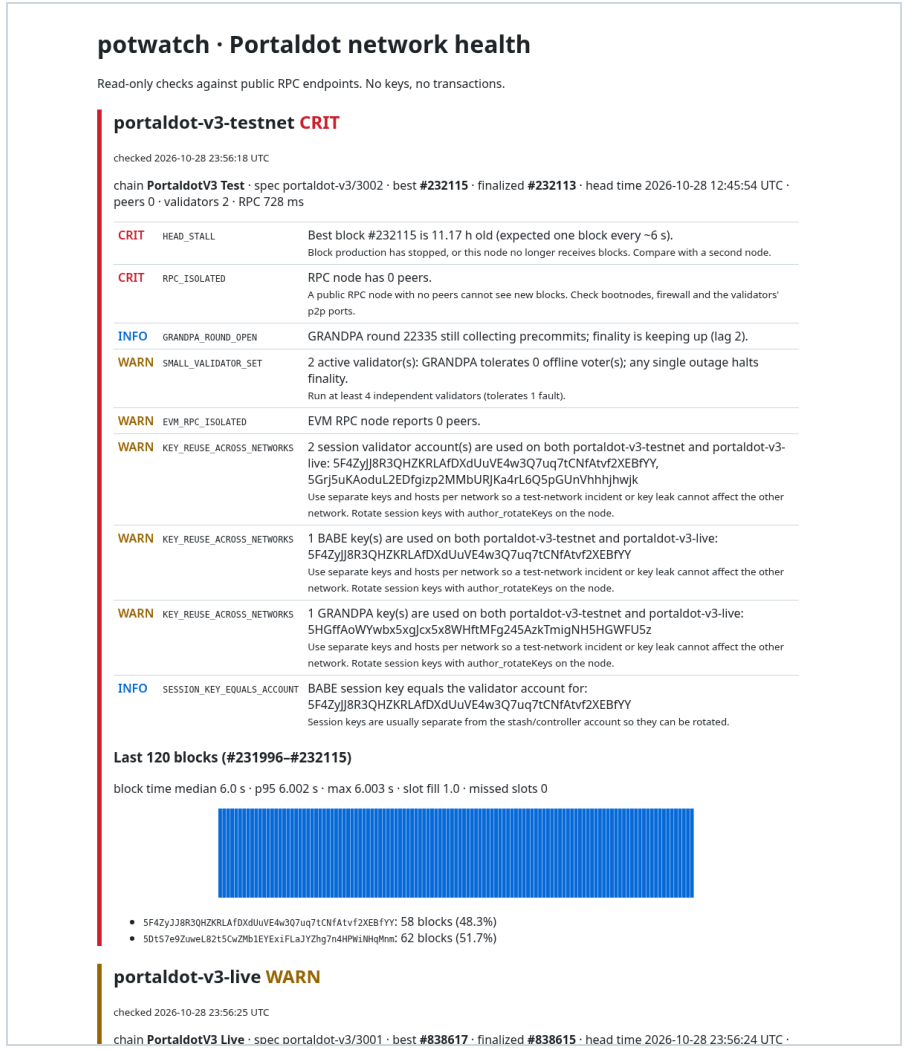
First real run: 2026-10-28, 23:56 UTC

Network	Verdict	What potwatch found
PortaldotV3 Test	**CRIT**	Head stuck at #232115 for 11.2 h; RPC and EVM nodes have 0 peers
PortaldotV3 Live	**WARN**	1 peer; 2 validators (GRANDPA tolerates 0 faults)
Both	**WARN**	The same 2 validator accounts, 1 BABE key and 1 GRANDPA key are active on both networks

Before the stall, the 120 blocks were perfectly regular (6.0 s median, 58/62 author split). Production stopped abruptly, with no slow decline. That points to a network or host event, not a failing validator.

Observations from public endpoints at one point in time; not a security audit.

The report



potwatch · Portaldot network health

Read-only checks against public RPC endpoints. No keys, no transactions.

portaldot-v3-testnet CRIT

checked 2026-10-28 23:56:18 UTC

chain **PortaldotV3 Test** · spec portaldot-v3/3002 · best **#232115** · finalized **#232113** · head time 2026-10-28 12:45:54 UTC · peers 0 · validators 2 · RPC 728 ms

CRIT	HEAD_STALL	Best block #232115 is 11.17 h old (expected one block every ~6 s). Block production has stopped, or this node no longer receives blocks. Compare with a second node.
CRIT	RPC_ISOLATED	RPC node has 0 peers. A public RPC node with no peers cannot see new blocks. Check bootnodes, firewall and the validators' p2p ports.
INFO	GRANDPA_ROUND_OPEN	GRANDPA round 22335 still collecting precommits; finality is keeping up (lag 2).
WARN	SMALL_VALIDATOR_SET	2 active validator(s): GRANDPA tolerates 0 offline voter(s); any single outage halts finality. Run at least 4 independent validators (tolerates 1 fault).
WARN	EVM_RPC_ISOLATED	EVM RPC node reports 0 peers.
WARN	KEY_REUSE_ACROSS_NETWORKS	2 session validator account(s) are used on both portaldot-v3-testnet and portaldot-v3-live: 5F4ZyJJ8R3QHZKRLAfDXdUuVE4w3Q7uq7tCNfAtvf2XEBfYY, 5Grj5uKAoduL2EDfgizp2MMbURJKa4rL6Q5pGUnVhhhjhwjk Use separate keys and hosts per network so a test-network incident or key leak cannot affect the other network. Rotate session keys with author_rotateKeys on the node.
WARN	KEY_REUSE_ACROSS_NETWORKS	1 BABE key(s) are used on both portaldot-v3-testnet and portaldot-v3-live: 5F4ZyJJ8R3QHZKRLAfDXdUuVE4w3Q7uq7tCNfAtvf2XEBfYY Use separate keys and hosts per network so a test-network incident or key leak cannot affect the other network. Rotate session keys with author_rotateKeys on the node.
WARN	KEY_REUSE_ACROSS_NETWORKS	1 GRANDPA key(s) are used on both portaldot-v3-testnet and portaldot-v3-live: 5HGffAoWYwbx5xgJcx5x8WHftMFg245AzkTmigNH5HGWFU5z Use separate keys and hosts per network so a test-network incident or key leak cannot affect the other network. Rotate session keys with author_rotateKeys on the node.
INFO	SESSION_KEY_EQUALS_ACCOUNT	BABE session key equals the validator account for: 5F4ZyJJ8R3QHZKRLAfDXdUuVE4w3Q7uq7tCNfAtvf2XEBfYY Session keys are usually separate from the stash/controller account so they can be rotated.

Last 120 blocks (#231996–#232115)

block time median 6.0 s · p95 6.002 s · max 6.003 s · slot fill 1.0 · missed slots 0

- 5F4ZyJJ8R3QHZKRLAfDXdUuVE4w3Q7uq7tCNfAtvf2XEBfYY: 58 blocks (48.3%)
- 5Dt57e9ZuweL82t5CwZMb1EYExiFLaJYZhg7n4HPWiNHqMnm: 62 blocks (51.7%)

portaldot-v3-live WARN

checked 2026-10-28 23:56:25 UTC

chain **PortaldotV3 Live** · spec portaldot-v3/3001 · best **#838617** · finalized **#838615** · head time 2026-10-28 23:56:24 UTC ·

examples/2026-10-29/report.html

On-chain attestations on Portaldot EVM

- `HealthAttestation.sol` is an append-only log of report digests (sha256 of canonical JSON) with network, best/finalized block, timestamp and severity
- Anyone can re-hash the published `report.json` and compare it with `latest(network)`
- This gives a tamper-evident uptime history for networks and node operators, a building block for operator reputation and SLAs
- No funds held; only allow-listed reporters; stale reports rejected; 14 keyless EVM tests

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Testnet deployment: contract address and tx hash added after deployment
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Why Portaldot, and what comes next

Why Portaldot

- V3 combines Substrate consensus with an EVM. potwatch watches both, and uses the EVM to make the results verifiable.
- Small validator sets make early warning especially valuable right now.

Roadmap

- Hosted public status page plus periodic on-chain attestations
- Per-operator view: authored blocks, missed slots, GRANDPA participation
- Alerts by webhook, Discord or Telegram
- WASM (ink!) version of the attestation contract